Engineering Mathematics - 500+ One-Liners

Coal India Limited - Management Trainee (Systems) | CBT Preparation

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Table of Contents

Engineering Mathematics - 500+ One-Liners 4

1. Mathematical Logic 4

Propositional Logic 4
Tautology, Contradiction, Contingency 4
Logical Equivalences 4
Normal Forms 4
Rules of Inference 4
Predicate Logic 5

2. Set Theory and Relations 5

Sets and Power Sets 5
Inclusion-Exclusion 5
Relations 5
Functions 5
Countability 5

3. Partial Orders and Lattices 5

4. Algebraic Structures 6

5. Graph Theory 6

6. Combinatorics 6

7. Matrices and Determinants 7

8. System of Linear Equations 7

9. Eigenvalues and Eigenvectors 7

10. LU Decomposition 8

11. Limits, Continuity, Differentiability 8

12. Applications of Derivatives 8

13. Integration 8

14. Probability 9

15. Probability Distributions **9**

16. Statistics **9**

17. Numerical Solutions of Equations **10**

18. Numerical Integration **10**

19. Numerical Solution of ODEs **10**

20. Exam Focus Rapid-Fire **10**

21. Extended Rapid-Fire: Logic and Sets **11**

22. Extended Rapid-Fire: Lattices and Algebra **11**

23. Extended Rapid-Fire: Graphs **11**

24. Extended Rapid-Fire: Combinatorics **12**

25. Extended Rapid-Fire: Linear Algebra **12**

26. Extended Rapid-Fire: Calculus **13**

27. Extended Rapid-Fire: Probability **13**

28. Extended Rapid-Fire: Distributions and Statistics **14**

29. Extended Rapid-Fire: Numerical Methods **14**

30. Extended Rapid-Fire: Cross-Topic Mix **15**

31. Extended Rapid-Fire: More Logic and Proof **15**

32. Extended Rapid-Fire: More Sets and Relations **15**

33. Extended Rapid-Fire: More Linear Algebra and Numerical **16**

34. Extended Rapid-Fire: More Calculus and Series **16**

35. Extended Rapid-Fire: More Probability and Stats **16**

36. Extended Rapid-Fire: Final Mixed Set **17**

37. Bonus One-Liners **17**

Engineering Mathematics - 500+ One-Liners

- This rapid-revision book packs **500+** one-liners across **5** pillars: **Discrete Math**, **Linear Algebra**, **Calculus**, **Probability/Statistics**, and **Numerical Methods**.

1. Mathematical Logic

Propositional Logic

- There are **5** basic connectives: **NOT (¬)**, **AND (∧)**, **OR (∨)**, **implication (→)**, and **biconditional (↔)**.
- A truth table for **n** variables has exactly **2^n** rows.
- **Implication p→q** is false in only **1** case: when **p is true and q is false**.
- **Conjunction p∧q** is true in only **1** of **4** rows: when **both** are true.
- **Disjunction p∨q** is false in only **1** row: when **both** are false.
- **Biconditional p↔q** is true in exactly **2** rows: when **both truth values match**.
- The **XOR** operator is true in exactly **2** of **4** rows: when **operands differ**.
- The implication equivalence is **p→q ≡ ¬p∨q**, using **1** negation and **1** OR.
- The **1** equivalent form of an implication is its **contrapositive ¬q→¬p**.
- There are **2** non-equivalent transforms of p→q: its **converse q→p** and **inverse ¬p→¬q**.
- A **vacuously true** implication occurs whenever the premise is **false**, giving value **true**.

Tautology, Contradiction, Contingency

- There are **3** classifications of a formula: **tautology**, **contradiction**, and **contingency**.
- A **tautology** is true in all **2^n** rows, e.g., **p∨¬p**.
- A **contradiction** is false in all rows, e.g., **p∧¬p**.
- A **contingency** has at least **1** true and **1** false row.
- A formula is **satisfiable** if it is true in at least **1** row.

Logical Equivalences

- **De Morgan's** 2 laws are **¬(p∧q)≡¬p∨¬q** and **¬(p∨q)≡¬p∧¬q**.
- The **2** identity laws are **p∧T≡p** and **p∨F≡p**.
- The **2** domination laws are **p∨T≡T** and **p∧F≡F**.
- The **2** idempotent laws are **p∨p≡p** and **p∧p≡p**.
- The **absorption law** states **p∨(p∧q)≡p**.

Normal Forms

- There are **2** canonical normal forms: **CNF** (AND of ORs, product of sums) and **DNF** (OR of ANDs, sum of products).
- **DNF** is a **sum of products (SOP)**; **CNF** is a **product of sums (POS)**.

Rules of Inference

- **Modus Ponens** uses **2** premises **p** and **p→q** to infer **q**.
- **Modus Tollens** uses **¬q** and **p→q** to infer **¬p**.
- **Hypothetical syllogism** chains **p→q** and **q→r** into **p→r**.
- **Disjunctive syllogism** uses **p∨q** and **¬p** to infer **q**.
- There are **2** classic fallacies: **affirming the consequent** and **denying the antecedent**.

Predicate Logic

- There are **2** quantifiers: **universal** $\forall$ ("for all") and **existential** $\exists$ ("there exists").
- Negation flips quantifiers: $\neg\forall x\ P \equiv \exists x\ \neg P$ and $\neg\exists x\ P \equiv \forall x\ \neg P$.
- Quantifier order matters: $\forall x\exists y$ is weaker than $\exists y\forall x$.

2. Set Theory and Relations

Sets and Power Sets

- A set of **n** elements has 2^n subsets and $2^n - 1$ proper subsets.
- The **power set** of an n-element set has 2^n members.
- The **empty set** $\varnothing$ is a subset of **every** set and has exactly **1** subset (itself).
- The set $\{\varnothing\}$ has exactly **1** element, distinct from $\varnothing$.
- There are **4** core set operations: **union**, **intersection**, **difference**, and **complement**.
- The number of size-r subsets of an n-set is $C(n,r)$.

Inclusion-Exclusion

- The 2-set formula is $|A\cup B| = |A|+|B|-|A\cap B|$.
- The 3-set formula adds **3** singles, subtracts **3** pairs, and adds **1** triple.
- The Cartesian product satisfies $|A\times B| = |A|\cdot|B|$.

Relations

- A relation on n elements is any of $2^{(n^2)}$ subsets of $A\times A$.
- There are **4** key relation properties: **reflexive**, **symmetric**, **antisymmetric**, **transitive**.
- The count of **reflexive** relations on n elements is $2^{(n^2-n)}$.
- The count of **symmetric** relations on n elements is $2^{(n(n+1)/2)}$.
- An **equivalence relation** satisfies **3** properties: **reflexive**, **symmetric**, **transitive**.
- An equivalence relation **partitions** a set into **disjoint** equivalence classes.
- The number of equivalence relations on n elements equals the **Bell number** $B(n)$.
- A **partial order** satisfies **3** properties: **reflexive**, **antisymmetric**, **transitive**.

Functions

- There are **3** function types: **injective** (one-one), **surjective** (onto), **bijective** (both).
- Total functions from an m-set to an n-set number n^m.
- Injective functions from m to n ($m\le n$) number $n!/(n-m)!$.
- A **bijection** is the only type with an **inverse**.
- Composition **(f■g)(x)=f(g(x))** applies **g first**.

Countability

- **■ and ■** are **countable**; **■** is **uncountable**.

3. Partial Orders and Lattices

- A **POSET** requires **3** properties: **reflexive**, **antisymmetric**, **transitive**.
- A **lattice** requires that every pair have both a **LUB (join)** and **GLB (meet)**.

- A **bounded lattice** has **2** special elements: a least **0** and greatest **1**.
- A **Boolean algebra** is a lattice with **3** traits: **bounded**, **distributive**, **complemented**.
- Every **chain** (total order) is automatically a **distributive** lattice.
- The **LUB** is the **least upper bound**; the **GLB** is the **greatest lower bound**.

4. Algebraic Structures

- A **semigroup** needs **2** properties: **closure** and **associativity**.
- A **monoid** adds **1** property to a semigroup: an **identity element**.
- A **group** needs **4** properties: **closure**, **associativity**, **identity**, **inverse**.
- An **abelian group** adds **1** property: **commutativity**.
- **Lagrange's theorem** says a subgroup's order **divides** the group's order.
- Every **cyclic group** is **abelian** and generated by **1** element.
- The **order of an element a** is the smallest k with **a^k = e**.

5. Graph Theory

- The **handshaking lemma** states Σ **deg(v) = 2|E|**.
- The number of **odd-degree** vertices in any graph is always **even**.
- A **complete graph Kn** has **n(n–1)/2** edges.
- A **complete bipartite Kmn** has **m·n** edges.
- A **tree** on n vertices has exactly **n–1** edges.
- A graph is **bipartite** iff it has **no odd-length cycle**.
- An **Eulerian circuit** exists iff the graph is connected and all vertices have **even** degree.
- An **Eulerian path** exists iff exactly **0 or 2** vertices have **odd** degree.
- A **Hamiltonian** cycle visits every **vertex** once and has **no** simple necessary-sufficient condition.
- **Euler's formula** for connected planar graphs is **V – E + F = 2**.
- A simple planar graph satisfies **E ≤ 3V – 6**; bipartite planar satisfies **E ≤ 2V – 4**.
- The **2** smallest non-planar graphs are **K5** and **K3,3** (Kuratowski).
- The **chromatic number** of **Kn** is **n**, of a bipartite graph is **2**.
- An **odd cycle** has chromatic number **3**; an **even cycle** has **2**.
- The **Four Color Theorem** says every planar graph needs at most **4** colors.
- The number of spanning trees of **Kn** is **n^(n–2)** (Cayley).
- Adding one edge to a tree creates exactly **1** cycle.

6. Combinatorics

- The **2** basic counting rules are the **sum rule** (+) and **product rule** (×).
- A **permutation P(n,r) = n!/(n–r)!** counts arrangements where **order matters**.
- A **combination C(n,r) = n!/(r!(n–r)!)** counts selections where **order does not matter**.
- **Circular permutations** of n objects number **(n–1)!**.
- The **pigeonhole principle** guarantees some box holds at least **■n/k■** items.
- The **binomial theorem** expands **(x+y)^n = Σ C(n,k) x^(n–k) y^k**.
- The sum of binomial coefficients for power n is **2^n**.
- A linear homogeneous recurrence is solved via its **characteristic equation**.
- Distinct roots give **a_n = A r1^n + B r2^n**; a repeated root gives **(A+Bn) r^n**.

- The number of **derangements** is **D(n) = n! Σ(−1)^k/k!**.
- The basic generating function is **1/(1−x) = Σ x^n**.

7. Matrices and Determinants

- The **identity matrix I** satisfies **AI = IA = A** with **1**s on the diagonal.
- A **symmetric** matrix satisfies **A = A■**; a **skew-symmetric** satisfies **A = −A■**.
- A skew-symmetric matrix has all **0**s on its diagonal.
- An odd-order skew-symmetric matrix has determinant **0**.
- An **orthogonal** matrix satisfies **AA■ = I**, so **A■^{1} = A■** and **det = ±1**.
- Matrix multiplication is **associative** and **distributive** but **not commutative**.
- The product transpose rule is **(AB)■ = B■ A■** (reversed order).
- The product inverse rule is **(AB)■^{1} = B■^{1} A■^{1}** (reversed order).
- Naive n×n matrix multiplication costs **$O(n^3)$**; Strassen costs **O(n^2.81)**.
- A 2×2 determinant equals **ad − bc**.
- **det(A■) = det(A)** and **det(AB) = det(A)·det(B)**.
- Scaling an n×n matrix gives **det(kA) = k^n det(A)**.
- Swapping **2** rows multiplies the determinant by **−1**.
- A matrix with **2** identical rows has determinant **0**.
- The inverse formula is **A■^{1} = adj(A)/det(A)**, valid iff **det ≠ 0**.
- The product **A·adj(A) = det(A)·I**.
- For an n×n matrix, **det(adj A) = det(A)^(n−1)**.
- The **rank** of a matrix is the number of **linearly independent rows** (= columns).
- Rank equals the number of **non-zero rows** in row-echelon form.
- The only matrix with rank **0** is the **zero matrix**.
- There are **3** elementary row operations: **row swap**, **scaling**, **add-multiple-of-row**.
- Elementary row operations preserve **rank** and the **solution set**.

8. System of Linear Equations

- A **homogeneous** system **Ax=0** always has the **trivial** solution.
- For **Ax=b**, the system is **inconsistent** when **rank(A) < rank([A|b])**.
- A **unique** solution exists when **rank(A) = rank([A|b]) = n**.
- **Infinite** solutions occur when **rank(A) = rank([A|b]) < n**, with **n−rank** free variables.
- Homogeneous **Ax=0** has non-trivial solutions iff **det(A) = 0**.
- **Gaussian elimination** reduces to **upper triangular** form then back-substitutes, costing **$O(n^3)$**.
- **Cramer's rule** gives **x_i = det(A_i)/det(A)** and fails when **det(A)=0**.

9. Eigenvalues and Eigenvectors

- The eigenvalue equation is **Av = λv** with **v ≠ 0**.
- Eigenvalues solve the **characteristic equation det(A − λI) = 0**.
- The **sum** of eigenvalues equals the **trace** of A.
- The **product** of eigenvalues equals the **determinant** of A.
- A **triangular** matrix's eigenvalues are its **diagonal entries**.
- **Symmetric** matrices have all **real** eigenvalues.

- Eigenvalues of **A■^{1}** are **1/λ**; of **A^k** are **λ^k**.
- A **singular** matrix has **0** as an eigenvalue.
- The **Cayley-Hamilton theorem** says every matrix satisfies its own **characteristic equation**.
- Diagonalization writes **A = PDP■^{1}**, valid iff A has **n** independent eigenvectors.

10. LU Decomposition

- **LU decomposition** factors **A = LU** into **lower** and **upper** triangular matrices.
- **Doolittle** puts unit diagonal on **L**; **Crout** puts it on **U**.
- LU solves **Ly=b** by forward substitution then **Ux=y** by back substitution.
- LU factorization costs **$O(n^3)$**; each subsequent solve costs **$O(n^2)$**.
- LU is preferred for **multiple right-hand sides** since L,U are reused.

11. Limits, Continuity, Differentiability

- The standard limit **lim (sin x)/x = 1** as x→0.
- The standard limit **lim (1–cos x)/x^2 = 1/2** as x→0.
- The standard limit **lim (e^x – 1)/x = 1** as x→0.
- The standard limit **lim (1 + 1/n)^n = e** as n→∞.
- **L'Hopital's rule** applies only to **0/0** or **∞/∞** forms: **lim f/g = lim f'/g'**.
- f is **continuous** at a iff **lim f(x) = f(a)**.
- **Differentiability** implies **continuity**, but not the reverse.
- The function **|x|** is continuous but **not differentiable** at **x = 0**.
- The **product rule** is **(uv)' = u'v + uv'**.
- The **quotient rule** is **(u/v)' = (u'v – uv')/v^2**.
- The **chain rule** is **d/dx f(g(x)) = f'(g(x))·g'(x)**.
- The power derivative is **d/dx x■ = n x^(n–1)**.

12. Applications of Derivatives

- **Critical points** occur where **f'(x) = 0** or f' is undefined.
- The **second derivative test**: **f''>0** is a minimum, **f''<0** a maximum, **f''=0** inconclusive.
- An **inflection point** occurs where **f''=0** and concavity changes.
- **Global extrema** on [a,b] require checking critical points **and endpoints**.
- **Rolle's theorem** guarantees **f'(c)=0** when **f(a)=f(b)**.
- **Lagrange's MVT** guarantees **f'(c) = (f(b)–f(a))/(b–a)**.
- **Rolle's** is the special case of **Lagrange's** when **f(a)=f(b)**.

13. Integration

- The power integral is **∫x■ dx = x^(n+1)/(n+1) + C** for **n ≠ –1**.
- The reciprocal integral is **∫(1/x)dx = ln|x| + C**.
- **Integration by parts** is **∫u dv = uv – ∫v du**.
- The u-choice order in by-parts follows **ILATE**: Inverse, Log, Algebraic, Trig, Exponential.
- The **Fundamental Theorem** gives **∫_a^b f = F(b) – F(a)**.
- The **king property** is **∫_0^a f(x)dx = ∫_0^a f(a–x)dx**.

- Area under a curve uses **$\int_a^b |f(x)|\,dx$** to avoid sign cancellation.

14. Probability

- The probability axioms require **$0 \le P(A) \le 1$** and **$P(S) = 1$**.
- The general addition rule is **$P(A\cup B) = P(A)+P(B)-P(A\cap B)$**.
- **Conditional probability** is **$P(A|B) = P(A\cap B)/P(B)$**.
- The **multiplication rule** is **$P(A\cap B) = P(A|B)\cdot P(B)$**.
- **Independent** events satisfy **$P(A\cap B) = P(A)\cdot P(B)$**.
- **Mutually exclusive** non-zero events are always **dependent**.
- **Total probability** is **$P(A) = \Sigma\, P(A|B_i)P(B_i)$**.
- **Bayes' theorem** computes **$P(B_i|A) = P(A|B_i)P(B_i)/\Sigma P(A|B_j)P(B_j)$**.
- There are **2** random variable types: **discrete** (PMF) and **continuous** (PDF).
- A **PMF** sums to **1**; a **PDF** integrates to **1**.
- For a **continuous** RV, **$P(X = a) = 0$**.
- The **CDF** is **$F(x) = P(X \le x)$**, non-decreasing from **0** to **1**.
- **Expectation** is **$E[X] = \Sigma\, x\, P(x)$** or **$\int x\, f(x)\, dx$**.
- **Variance** is **$Var(X) = E[X^2] - (E[X])^2$**.
- Linear transform variance is **$Var(aX+b) = a^2\, Var(X)$**.
- Expectation is linear: **$E[X+Y] = E[X]+E[Y]$** always.
- **$Var(X+Y) = Var(X)+Var(Y)$** only when X,Y are **independent**.

15. Probability Distributions

- The **binomial** PMF is **$C(n,k)\, p^k\, (1-p)^{(n-k)}$** with mean **np** and variance **$np(1-p)$**.
- The **Poisson** PMF is **$e^{(-\lambda)}\, \lambda^k / k!$** with mean = variance = λ.
- Poisson approximates binomial when **n large, p small, $np = \lambda$**.
- The **continuous uniform** on [a,b] has mean **$(a+b)/2$** and variance **$(b-a)^2/12$**.
- The **normal** distribution has parameters mean μ and variance **σ^2**.
- The **standard normal $Z = (X-\mu)/\sigma$** has mean **0** and variance **1**.
- The **empirical rule** gives **68%**, **95%**, **99.7%** within 1, 2, 3 standard deviations.
- The **exponential** PDF is **$\lambda e^{(-\lambda x)}$** with mean **$1/\lambda$** and variance **$1/\lambda^2$**.
- The exponential distribution is **memoryless**: **$P(X>s+t|X>s)=P(X>t)$**.
- A normal distribution is **symmetric** with **mean = median = mode**.

16. Statistics

- The **3** central tendency measures are **mean**, **median**, and **mode**.
- The **mean** is sensitive to outliers; the **median** is robust.
- The empirical mode relation is **Mode = 3·Median – 2·Mean**.
- **Population variance** divides by **N**; **sample variance** divides by **n–1**.
- The **range** is **max – min**; **standard deviation** is **$\sqrt{\text{variance}}$**.
- The **coefficient of variation** is σ/μ.
- **Pearson's r** lies in **[–1, 1]** and equals **$cov(X,Y)/(\sigma_x \sigma_y)$**.
- **r = 0** indicates no **linear** correlation.
- The regression slope is **$b = cov(X,Y)/Var(X)$**.

- The correlation equals the **geometric mean** of the **2** regression coefficients: **$r = \sqrt{(bxy \cdot byx)}$**.

17. Numerical Solutions of Equations

- The **bisection** method needs **$f(a) \cdot f(b) < 0$** and has **linear** convergence (order **1**).
- Bisection error after n steps is **$\leq (b-a)/2^n$**.
- **Newton-Raphson** iterates **$x - f(x)/f'(x)$** with **quadratic** convergence (order **2**).
- Newton-Raphson fails when **$f'(x) = 0$**.
- The **secant** method has convergence order **≈1.618**.
- **Regula-falsi** keeps a **bracketing** interval, guaranteeing convergence.
- **Bisection** and **regula-falsi** are the **2** bracketing methods that always converge.
- **Gauss elimination** is a **direct** method costing **$O(n^3)$**.
- **Gauss-Seidel** converges when the matrix is **diagonally dominant**.
- **Gauss-Seidel** uses **updated** values immediately, unlike **Jacobi**.
- **Absolute error** is **|true − approx|**; **relative error** divides by **|true|**.

18. Numerical Integration

- The **trapezoidal rule** has error **$O(h^2)$** and is exact for degree **1** polynomials.
- **Simpson's 1/3 rule** has error **O(h■)** and is exact for degree **3** polynomials.
- **Simpson's 1/3** requires an **even** number of subintervals.
- The trapezoidal formula is **$(h/2)[f_0 + 2\Sigma f_{mid} + f_n]$**.
- Simpson's 1/3 weights are **1, 4, 2, 4, ..., 1** scaled by **h/3**.
- **Simpson's 3/8 rule** also has error **O(h■)** and is exact to degree **3**.

19. Numerical Solution of ODEs

- **Euler's method** is **$y_{(n+1)} = y_n + h \cdot f(x_n, y_n)$**, first-order with error **$O(h)$**.
- **RK4** is **$y_{(n+1)} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$**, order **4**.
- RK4 evaluates the slope **4** times per step, weighting middle slopes by **2**.
- **Multi-step** methods like **Adams-Bashforth** use **several** previous points.
- A **predictor-corrector** pairs an **explicit** predictor with an **implicit** corrector.

20. Exam Focus Rapid-Fire

- Test logical equivalence using **truth tables** or the **De Morgan / distributive** laws.
- Classify relations: **RST** gives an **equivalence**, **RAT** gives a **partial order**.
- Confirm a lattice by checking **LUB and GLB** exist for **all** pairs.
- Decide counting type: **order matters** is permutation, otherwise **combination**.
- Solve recurrences via the **characteristic equation**, using **$(A+Bn)r^n$** for repeated roots.
- Use **row-echelon** form to find rank and **A■¹ = adj(A)/det(A)** for inverse.
- Apply **trace = $\Sigma\lambda$** and **det = $\Pi\lambda$** as quick eigenvalue checks.
- Use **Bayes** when conditioning must be **reversed**.
- Remember **Newton-Raphson order 2**, **trapezoidal $O(h^2)$**, **Simpson O(h■)**.
- Apply **handshaking $\Sigma deg=2E$** and **Euler $V-E+F=2$** for graph counting questions.

21. Extended Rapid-Fire: Logic and Sets

- The **commutative** laws hold for $\wedge$ and $\vee$ but **not** for $\rightarrow$.
- The **associative** law lets $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ regroup freely.
- The **distributive** law gives $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- The **2** negation laws are $\neg\neg p \equiv p$ (double negation) and $\neg T \equiv F$.
- A formula with **k** clauses in CNF is **satisfiable** if all clauses can be **simultaneously true**.
- The **addition** rule infers $p \vee q$ from a single **p**.
- The **simplification** rule infers **p** from $p \wedge q$.
- The **conjunction** rule infers $p \wedge q$ from **2** premises **p** and **q**.
- The **resolution** rule combines $p \vee q$ and $\neg p \vee r$ into $q \vee r$.
- A **predicate** becomes a proposition after **quantification** or **value assignment**.
- The statement "**some**" maps to $\exists$; "**every/all**" maps to $\forall$.
- Negating "all swans are white" gives "**there exists a swan that is not white**".
- The set **difference** satisfies $A - B = A \cap B'$.
- **Symmetric difference A■B =** $(A-B) \cup (B-A)$ contains elements in exactly **1** set.
- Two sets are **disjoint** when $A \cap B = \varnothing$.
- The **universal set U** satisfies $A \cup A' = U$ and $A \cap A' = \varnothing$.
- **De Morgan** for sets: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.
- The number of **onto** functions from a 3-set to a 2-set is **6**.
- A relation that is both **symmetric and antisymmetric** contains only **diagonal** pairs.
- An **irreflexive** relation contains **no** pair of the form **(a,a)**.
- A **total (linear) order** adds **comparability** to a partial order.
- The **transitive closure** adds the minimum pairs to make a relation **transitive**.
- A function from an n-set to itself that is injective is automatically **surjective** (and bijective).
- The number of **bijections** on an n-set is **n!**.

22. Extended Rapid-Fire: Lattices and Algebra

- The **join** of two elements is their **LUB**; the **meet** is their **GLB**.
- In a **distributive** lattice, complements (if they exist) are **unique**.
- A **sublattice** is closed under both **meet** and **join**.
- The divisibility relation on positive integers forms a **lattice** with meet **gcd** and join **lcm**.
- The power set under $\subseteq$ forms a **Boolean lattice** with **2^n** elements.
- The **identity** element of addition is **0**; of multiplication is **1**.
- A **ring** has **2** operations with addition forming an **abelian group**.
- A **field** requires every **non-zero** element to have a multiplicative **inverse**.
- The set of integers under addition is an **abelian group** but under multiplication only a **monoid**.
- The **order** of the group **(Zn, +)** is **n**.
- A group with **prime** order is always **cyclic**.
- The **identity** of a group is **unique**, and each element has a **unique** inverse.

23. Extended Rapid-Fire: Graphs

- A **walk** allows repeats; a **trail** repeats no edge; a **path** repeats no vertex.
- A graph with **n** vertices can have at most $n(n-1)/2$ edges if simple.

- A **regular** graph of degree r on n vertices has **$nr/2$** edges.
- The **minimum** number of edges in a connected graph on n vertices is **$n-1$**.
- A **forest** with k components and n vertices has **$n-k$** edges.
- A **cut vertex** disconnects the graph when removed.
- A **bridge** is an edge whose removal increases the number of **components**.
- The **complement** of a graph on n vertices and e edges has **$n(n-1)/2 - e$** edges.
- A **self-complementary** graph on n vertices needs **$n \equiv 0$ or $1 \pmod 4$**.
- A **tournament** is a complete graph with **directed** edges.
- The **adjacency matrix** of an undirected graph is **symmetric**.
- The **(i,j)** entry of **A^k** counts walks of length **k** from i to j.
- A **tree** with n vertices has exactly **$n-1$** edges and **0** cycles.
- A **binary tree** of height h has at most **$2^{(h+1)} - 1$** nodes.
- A graph is a **tree** iff it is **connected** and has **$n-1$** edges.
- The **degree** of every vertex in a cycle **C_n** is **2**.
- The **Petersen graph** is a famous **non-planar**, **3-regular** graph.
- A planar graph divides the plane into **F** regions counting the **outer** face.

24. Extended Rapid-Fire: Combinatorics

- Arranging n objects with repeats **$n_1, n_2, \ldots$** gives **$n!/(n_1!\, n_2!\, \ldots)$**.
- Selecting r from n **with repetition** (combinations) gives **$C(n+r-1, r)$**.
- The number of binary strings of length n is **2^n**.
- The number of n-bit strings with exactly k ones is **$C(n,k)$**.
- The **Catalan number $C_n = C(2n,n)/(n+1)$** counts balanced parentheses.
- The number of ways to seat n people around a circular table is **$(n-1)!$**.
- **Pascal's identity** is **$C(n,k) = C(n-1,k-1) + C(n-1,k)$**.
- The middle binomial coefficient **$C(n, n/2)$** is the **largest**.
- The **Fibonacci** recurrence is **$F_n = F_{(n-1)} + F_{(n-2)}$** with characteristic root **golden ratio ϕ**.
- The number of subsets of even size of an n-set is **$2^{(n-1)}$**.
- The **pigeonhole** principle guarantees a repeat among **$n+1$** items in **n** categories.
- A recurrence with characteristic roots **2 and 3** has solution **$A \cdot 2^n + B \cdot 3^n$**.

25. Extended Rapid-Fire: Linear Algebra

- A matrix is **singular** iff its determinant is **0**.
- The **trace** is the sum of **diagonal** entries and is **invariant** under similarity.
- The **transpose** satisfies **(A■)■ = A** and **(A+B)■ = A■+B■**.
- A **nilpotent** matrix has all eigenvalues **0**.
- An **idempotent** matrix satisfies **$A^2 = A$** with eigenvalues **0 or 1**.
- An **involutory** matrix satisfies **$A^2 = I$** with eigenvalues **± 1**.
- The rank of an **$m \times n$** matrix is at most **$\min(m,n)$**.
- The **nullity** plus the **rank** equals the number of **columns** (rank-nullity).
- A **diagonal** matrix's inverse inverts each **diagonal** entry.
- The determinant of a **diagonal/triangular** matrix is the **product** of diagonal entries.
- The number of **pivots** in row-echelon form equals the **rank**.
- **Similar** matrices share the same **eigenvalues**, **trace**, and **determinant**.

- A **positive definite** matrix has all **positive** eigenvalues.
- The **characteristic polynomial** of an n×n matrix has degree **n**.
- For a 2×2 matrix, eigenvalues solve **λ^2 – (trace)λ + det = 0**.
- A **real symmetric** matrix is always **diagonalizable** by an orthogonal matrix.
- The **condition number** measures sensitivity of **Ax=b** to perturbations.
- An **augmented matrix** appends the constant vector **b** as an extra column.

26. Extended Rapid-Fire: Calculus

- The **derivative** of **sin x** is **cos x**, of **cos x** is **–sin x**.
- The **derivative** of **tan x** is **$\sec^2 x$**.
- The **derivative** of **ln x** is **1/x**, of **e■** is **e■**.
- The **derivative** of **a■** is **a■ ln a**.
- The **nth derivative** of **e■** is still **e■**.
- A function is **increasing** where **f' > 0** and decreasing where **f' < 0**.
- A function is **concave up** where **f'' > 0**.
- The **tangent** line at x=a has slope **f'(a)**.
- The limit **lim (tan x)/x = 1** as x→0.
- The limit **lim (ln(1+x))/x = 1** as x→0.
- A **removable** discontinuity occurs when the limit **exists** but ≠ f(a).
- A **jump** discontinuity has unequal **left and right** limits.
- The **Squeeze theorem** bounds a function between **2** others with equal limits.
- **∫$\sec^2$x dx = tan x + C** and **∫(1/(1+x^2)) dx = arctan x + C**.
- **∫(1/√(1–x^2)) dx = arcsin x + C**.
- The integral **∫_0^∞ e^(–x) dx = 1**.
- A **definite integral** of an **odd** function over **[–a,a]** is **0**.
- A **definite integral** of an **even** function over **[–a,a]** is **2∫_0^a**.
- The **average value** of f on [a,b] is **(1/(b–a))∫_a^b f dx**.
- The **area between two curves** is **∫(top – bottom) dx**.
- An **improper integral** has an infinite **limit** or unbounded **integrand**.
- The **Gamma function** satisfies **Γ(n) = (n–1)!** for positive integers.

27. Extended Rapid-Fire: Probability

- The probability of the **complement** is **P(A') = 1 – P(A)**.
- The probability of the **impossible** event is **0**; of the **certain** event is **1**.
- For **equally likely** outcomes, **P(A) = favorable/total**.
- Rolling 2 dice has **36** outcomes; sum **7** has probability **6/36 = 1/6**.
- A standard deck has **52** cards, **4** suits, and **13** ranks.
- Drawing an **ace** from a deck has probability **4/52 = 1/13**.
- The **odds** in favor are **P(A) : P(A')**.
- **Mutually exclusive** events satisfy **P(A∩B) = 0**.
- **Pairwise independence** does **not** imply **mutual independence**.
- The **expected value** of a fair die is **3.5**.
- The **variance** of a single fair die is **35/12**.
- A **Bernoulli** trial has **2** outcomes with success probability **p**.

- The sum of **n** independent Bernoulli(p) trials is **Binomial(n,p)**.
- The **covariance** of independent variables is **0**.
- **Chebyshev's inequality** bounds **$P(|X-\mu| \geq k\sigma) \leq 1/k^2$**.
- The **law of large numbers** says the sample mean converges to μ.
- The **central limit theorem** says sums of many independent variables approach a **normal** distribution.

28. Extended Rapid-Fire: Distributions and Statistics

- The **geometric** distribution models trials until the **first success**, mean **$1/p$**.
- The **geometric** distribution is the only **discrete** memoryless distribution.
- The **hypergeometric** distribution models sampling **without replacement**.
- The **binomial** models sampling **with replacement** (independent trials).
- The **Poisson** models **rare** events over a fixed interval.
- The **standard normal** table gives **$P(Z \leq z)$** for the cumulative area.
- About **50%** of a normal distribution lies **below the mean**.
- The **mode** of a normal distribution equals its **mean**.
- A **right-skewed** distribution has **mean > median > mode**.
- A **left-skewed** distribution has **mean < median < mode**.
- The **interquartile range** is **Q3 – Q1**.
- The **median** is the **50th** percentile or **Q2**.
- **Variance** has units that are the **square** of the data units.
- **Standard deviation** has the **same** units as the data.
- A **perfect positive** correlation has **$r = 1$**; perfect negative has **$r = -1$**.
- The **regression** line always passes through the point **(x■, ■)**.
- The **product** of the two regression coefficients lies in **[0,1]** and equals **r^2**.
- **Skewness** measures asymmetry; **kurtosis** measures **tailedness**.

29. Extended Rapid-Fire: Numerical Methods

- **Bisection** guarantees convergence but is **slow** (order 1).
- **Newton-Raphson** converges fastest (order **2**) for a **simple** root.
- **Newton-Raphson** for **$\sqrt{a}$** iterates **$x = (x + a/x)/2$**.
- The **fixed-point** iteration converges if **$|g'(x)| < 1$**.
- The **secant** method needs **2** initial points but **no** derivative.
- **Jacobi** and **Gauss-Seidel** are the **2** classic iterative linear solvers.
- **Gauss-Seidel** typically converges **faster** than **Jacobi**.
- **Diagonal dominance** is a **sufficient** condition for iterative convergence.
- **Partial pivoting** swaps rows to put the **largest** pivot on the diagonal.
- **Trapezoidal** rule approximates the area with **trapezoids** (straight tops).
- **Simpson's 1/3** fits **parabolas** through groups of **3** points.
- **Simpson's 3/8** fits cubics through groups of **4** points.
- **Romberg integration** refines the trapezoidal rule via **Richardson extrapolation**.
- **Euler's method** accumulates **global** error of order **$O(h)$**.
- **RK4** accumulates global error of order **O(h■)**.
- The **step size h** controls the **accuracy** vs **cost** tradeoff.
- **Truncation error** comes from the **method**; **round-off error** from finite **precision**.

- **Newton's forward difference** interpolation suits **equally spaced** points near the start.
- **Lagrange interpolation** works for **unequally spaced** points.
- The **Newton-Raphson** convergence may **fail** near an **inflection** point or flat slope.

30. Extended Rapid-Fire: Cross-Topic Mix

- The **factorial** grows as **$O(n!)$**, faster than **exponential 2^n**.
- The **golden ratio $\phi \approx 1.618$** is the dominant Fibonacci root.
- **Euler's number $e \approx 2.71828$** is the base of natural logarithms.
- **$\pi \approx 3.14159$** relates a circle's circumference to its diameter.
- The **gcd** and **lcm** satisfy **$\gcd(a,b)\cdot\mathrm{lcm}(a,b) = a\cdot b$**.
- A number is divisible by **3** if its digit sum is divisible by **3**.
- The number of divisors of **$n = p^a q^b$** is **$(a+1)(b+1)$**.
- **Modular arithmetic** wraps values around a **modulus**.
- **Fermat's little theorem** states **$a^{(p-1)} \equiv 1 \pmod p$** for prime p.
- The sum **$1+2+\ldots+n$** equals **$n(n+1)/2$**.
- The sum of the first n squares is **$n(n+1)(2n+1)/6$**.
- The sum of a geometric series is **$a(1-r^n)/(1-r)$** for **$r \neq 1$**.
- An infinite geometric series converges to **$a/(1-r)$** when **$|r| < 1$**.
- The **harmonic series $\Sigma\ 1/n$ diverges**.
- **Logarithm** identity: **$\log(ab) = \log a + \log b$**.
- **Big-O** describes an **upper** bound; **Big-Ω** a **lower** bound; **Θ** a **tight** bound.

31. Extended Rapid-Fire: More Logic and Proof

- A **valid** argument has a true conclusion whenever all **premises** are true.
- A **sound** argument is valid **and** has all **true** premises.
- **Proof by contradiction** assumes the **negation** and derives a **false** statement.
- **Proof by contrapositive** proves **$\neg q \to \neg p$** instead of **$p \to q$**.
- **Mathematical induction** has **2** steps: the **base case** and the **inductive step**.
- **Strong induction** assumes the statement for **all** smaller cases.
- The **well-ordering principle** says every non-empty set of positive integers has a **least** element.
- A **counterexample** disproves a **universal** statement with a single instance.
- The **principle of explosion** says a **contradiction** implies **anything**.
- **Functional completeness**: **NAND** alone can express **all** Boolean functions.
- **NOR** alone is also **functionally complete**.
- There are **16** distinct binary Boolean functions of **2** variables.

32. Extended Rapid-Fire: More Sets and Relations

- The number of **antisymmetric** relations on n elements is **$2^n \cdot 3^{(n(n-1)/2)}$**.
- A relation can be neither **symmetric** nor **antisymmetric** simultaneously.
- The **composition** of two functions is associative: **(f■g)■h = f■(g■h)**.
- The **inverse** of a bijection f satisfies **f■f■¹ = identity**.
- A **finite** set is always **countable**.
- The set of all **subsets** of ■ is **uncountable**.

- **$|A \cup B \cup C|$** via inclusion-exclusion alternates **+,–,+** signs.
- The **floor** function maps reals to the **largest integer $\leq$ x**.
- The **ceiling** function maps reals to the **smallest integer $\geq$ x**.
- A **partial function** may be **undefined** on some domain elements.

33. Extended Rapid-Fire: More Linear Algebra and Numerical

- A **homogeneous** system with more **unknowns than equations** has **infinite** solutions.
- The **dot product** of orthogonal vectors is **0**.
- The **cross product** magnitude equals the **parallelogram** area.
- The **norm $||v|| = \sqrt{(v \cdot v)}$**.
- A set of vectors is **linearly independent** if the only zero combination is **trivial**.
- The **dimension** of a vector space is the size of its **basis**.
- The **column space** dimension equals the **rank**.
- **Eigenvectors** for distinct eigenvalues are **linearly independent**.
- The **spectral radius** is the **largest** absolute eigenvalue.
- A **stochastic** matrix has rows summing to **1** and largest eigenvalue **1**.
- The **power method** finds the **dominant** eigenvalue iteratively.
- **Cholesky** decomposition factors a **symmetric positive-definite** matrix as **LL■**.
- **QR** decomposition factors **A = QR** with **Q orthogonal** and **R upper triangular**.
- The **Frobenius norm** sums the **squares** of all entries.
- **Gauss-Jordan** elimination reduces to **reduced row-echelon** form.

34. Extended Rapid-Fire: More Calculus and Series

- The **Taylor series** expands f around a point using its **derivatives**.
- The **Maclaurin series** is a Taylor series centered at **0**.
- The Maclaurin series of **e■** is **Σ x■/n!**.
- The Maclaurin series of **sin x** has only **odd** powers.
- The Maclaurin series of **cos x** has only **even** powers.
- A **partial derivative** differentiates with respect to **one** variable, holding others fixed.
- The **gradient** points in the direction of **steepest ascent**.
- The **chain rule** for **dy/dx** via t is **(dy/dt)/(dx/dt)**.
- **L'Hopital** can be applied **repeatedly** while the form stays indeterminate.
- The **Wronskian** tests linear independence of **functions**.
- A **critical point** of a 2-variable function needs **both** partials zero.
- The **Hessian** determinant test classifies **2-variable** extrema.
- The **arc length** of a curve is **$\int\sqrt{(1 + (dy/dx)^2)}\, dx$**.
- The **volume of revolution** (disk method) is **$\int\pi[f(x)]^2\, dx$**.

35. Extended Rapid-Fire: More Probability and Stats

- The **multiplication rule** for **n** independent events multiplies all **P(Ai)**.
- The **birthday paradox** gives >**50%** chance of a shared birthday with just **23** people.
- A **joint** distribution describes **2 or more** random variables together.
- The **marginal** distribution sums/integrates out the **other** variables.

- **Conditional expectation E[X|Y]** is itself a **random variable**.
- The **moment generating function** encodes all **moments** of a distribution.
- The **first moment** about the origin is the **mean**.
- The **second central moment** is the **variance**.
- **Standardizing** a variable gives mean **0** and variance **1**.
- The **sample mean** is an **unbiased** estimator of the population mean.
- A **confidence interval** gives a range for a **parameter** at a stated confidence.
- **Type I error** rejects a true **null** hypothesis (false positive).
- **Type II error** fails to reject a false **null** hypothesis (false negative).
- The **p-value** is the probability of data at least as extreme under the **null**.
- The **normal** approximation to the binomial works when **np** and **n(1–p)** are ≥ **5**.

36. Extended Rapid-Fire: Final Mixed Set

- A **prime** number has exactly **2** divisors: **1** and itself.
- There are **25** primes below **100**.
- The number **2** is the only **even** prime.
- **Euclid's algorithm** computes the **gcd** by repeated **remainder**.
- The **base case** of recursion prevents **infinite** recursion.
- A **permutation matrix** has exactly one **1** per row and column.
- The **determinant** of a permutation matrix is **±1**.
- The **logarithm base change** formula is **log_b a = ln a / ln b**.
- **Stirling's approximation**: **n!** ≈ **√(2πn)(n/e)^n**.
- The **number of edges** in a complete bipartite graph **K3,3** is **9**.
- The **chromatic number** of a tree with at least one edge is **2**.
- A **spanning tree** of a graph with n vertices always has **n–1** edges.
- The **trapezoidal rule** with **1** interval uses just the **2** endpoints.
- **Simpson's rule** needs at least **2** subintervals (3 points).
- The **order** of convergence of Newton-Raphson at a **double** root drops to **1** (linear).
- A **Markov chain** transition matrix has each row summing to **1**.
- The **steady-state** vector of a Markov chain is the eigenvector for eigenvalue **1**.

37. Bonus One-Liners

- The **rank-nullity theorem** states **rank + nullity = number of columns**.
- A **scalar matrix** is a diagonal matrix with all equal **diagonal** entries.
- The **minor** of an entry is the determinant of the submatrix after deleting its **row and column**.
- The **cofactor** is a minor with sign **(–1)^(i+j)**.
- A **Vandermonde** determinant is non-zero when all **nodes are distinct**.
- The **conditional probability chain rule** multiplies **P(A1)P(A2|A1)P(A3|A1A2)**.
- A **discrete uniform** over **1..n** has mean **(n+1)/2**.
- The **Poisson** approximation improves as **n→∞** with **np** fixed.
- The **trapezoidal error** is proportional to the **second derivative f''**.
- **Simpson's error** is proportional to the **fourth derivative f■**.
- A **recurrence** of degree **k** needs **k** initial conditions.
- The **empty graph** on n vertices has **0** edges and **n** components.

- A **complete graph Kn** is **(n–1)**-regular.
- The **number of leaves** in a full binary tree with i internal nodes is **i+1**.
- The **determinant** of an orthogonal matrix is always **+1 or –1**.
- The **eigenvalues** of a projection matrix are **0 and 1**.